Example 5.28 A line makes positive intercepts on coordinate axes whose sum is 7 and it passes through $(-3,8)$. Find its equation.

Solution If a and b are the intercepts then $a+b=7$ or $b=7-a$

By intercept form $\quad \frac{x}{a}+\frac{y}{b}=1$

We have $\quad \frac{x}{a}+\frac{y}{7-a}=1$

As this line pass through the point $(-3,8)$, we have

$$\frac{-3}{a}+\frac{8}{7-a}=1 \quad \Rightarrow \quad -3(7-a)+8a=a(7-a)$$

$$-21+3a+8a=7a-a^2$$

So, $\quad a^2+4a-21=0$

Solving this equation $(a-3)(a+7)=0$

$$a=3 \quad \text{or} \quad a=-7$$

Since a is positive, we have $a=3$ and $b=7-a=7-3=4$.

Hence $\quad \frac{x}{3}+\frac{y}{4}=1$

Therefore, $4x+3y-12=0$ is the required equation.

Example 5.29 A circular garden is bounded by East Avenue and Cross Road. Cross Road intersects North Street at D and East Avenue at E. AD is tangential to the circular garden at $A(3,\ 10)$. Using the figure.

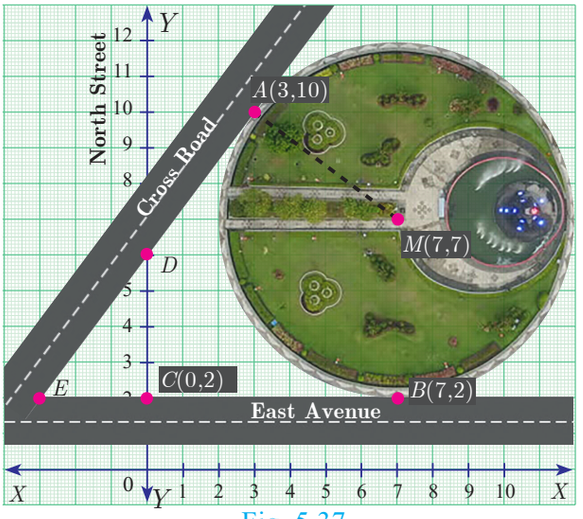


Fig. 5.37

(a) Find the equation of

(i) East Avenue.

(ii) North Street

(iii) Cross Road

(b) Where does the Cross Road intersect?

(i) North Street (ii) East Avenue

Solution (a) (i) East Avenue is the straight line joining $C(0,2)$ and $B(7,2)$. Thus the equation of East Avenue is obtained by using two-point form which is

$$\frac{y-2}{2-2}=\frac{x-0}{7-0}$$

$$\frac{y-2}{0}=\frac{x}{7} \quad \Rightarrow \quad y=2$$

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